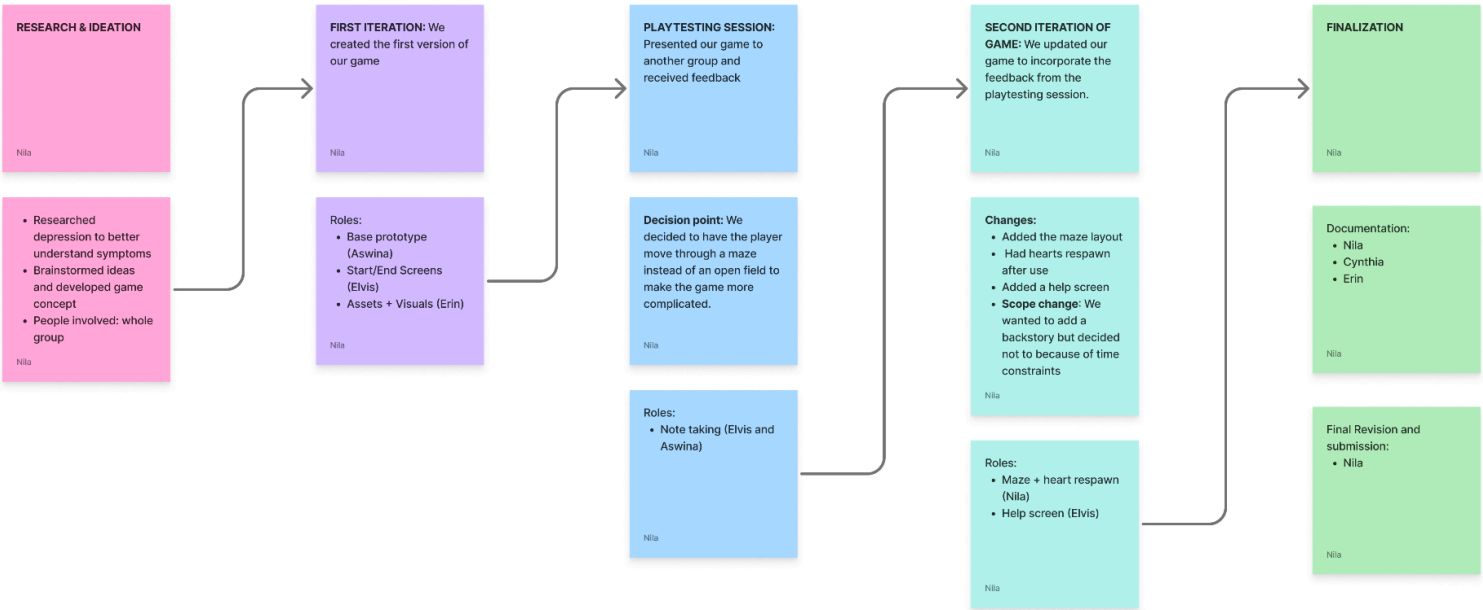
Process and Decision Documentation

Process Overview Visualization



Link to process visualization file:
https://www.figma.com/board/Gi54PyEUh0KFUx1MpuIIZH/Group-3B-Process-Visualization?node-id=0-1&t=d277rEiY2W2P7IBo-1

Project/Assignment Decisions

1. Maze Environment Design

What changed: The game environment was redesigned into a maze structure with walls and pathways instead of a simple open environment. Players must navigate through the maze to reach the objective.

Why the decision was made: The initial design lacked challenge and directional gameplay. Introducing a maze created a clearer game objective and spatial challenge, encouraging players to explore and think strategically about navigation. This also improved the visual structure of the game world and made the experience feel more complete.

Who was involved: The maze concept was discussed collectively by the group after our playtesting session. Layout ideas were proposed during group meetings, and Nila coded the final maze structure, while other members contributed suggestions for layout.

2. Heart Respawn Mechanic

What changed: A heart-based respawn system was introduced. When a player loses, they are able to respawn using a heart instead of immediately ending the game.

Why the decision was made: Early testing suggested that players found immediate failure frustrating. The heart system allows players to continue playing after mistakes, making the game more forgiving while still maintaining consequences for losing lives. This mechanic also improves engagement by encouraging players to keep exploring the maze.

Who was involved: This idea came from group discussions after internal testing. The respawn logic was implemented by Nila.

3. Help Overlay Screen

What changed: We added a help button that opened an overlay explaining how players move through the maze.

Why the decision was made: During testing, some players were unsure how to begin playing. Adding a help button in the corner helped reduce confusion and made the game easier for first-time players to understand.

Who was involved: This update was based on group feedback and playtesting observations. The text was rewritten collaboratively, and the overlay screen was coded by Elvis.

Role-Based Process Evidence

Aswina Sivaloganathan A2 Process Documentation

Name: Aswina Sivaloganathan
Role(s): Core Mechanics Programmer
Primary responsibility for this work: Create the base function that our group will work with and add to.

Goal of Work Session: Define the core structure of the game and establish a working base prototype that allows the player to move within the environment.

Tools, Resources, or Inputs Used

- p5.js
- Visual Studio Code
- GitHub

GenAI tools (if used): ChatGPT 5.2

Date Used: March 3rd 2026

Purpose of Use:
GenAI was used to help adapt an existing starter code from a previous side quest into a base structure that better aligned with our group's game idea. The goal was to create a functional starting point that our team could build on and expand throughout development.

Summary of Interaction:
I provided ChatGPT with starter code from a previous p5.js side quest and asked for assistance restructuring it into a base prototype that supports our game idea. The tool generated suggestions for organizing the code and implementing player movement. These suggestions were reviewed and modified before being integrated into the project.

Human Decision Point(s):
I decided to reuse code from a previous assignment as a starting point rather than building the structure entirely from scratch. After reviewing the AI-generated suggestions, she selected and adapted the relevant parts that supported the intended gameplay mechanics.

Integrity & Verification Note:
GenAI was used to assist with restructuring previously written code into a base template for the game. All final code adjustments and integration into the repository were completed manually.

Scope of GenAI Use:
GenAI was used to assist with restructuring previously written code into a base template for the game. All final code adjustments and integration into the repository were completed manually.

Limitations or Misfires:

Some of the generated code did not fully match the structure of the starter code from the previous side quest.

Verification & Judgement

The code was tested in the p5.js environment to confirm that the player movement and base structure worked correctly. After testing, the code was reviewed to ensure it was simple enough for the rest of the group to understand and expand upon. Only the parts that functioned correctly and supported the game concept were kept.

Limitations, Dead Ends, or Open Questions

At this stage, the prototype only includes the base structure and movement mechanics. Additional systems such as collision detection, UI elements, and environment assets still need to be integrated by other team members. Further testing will also be needed to ensure all systems work together smoothly.

Appendix: https://chatgpt.com/share/69b220e9-5f80-800f-b95f-52f4d8a5f157

Elvis Maciejewski A2

Name: Elvis Maciejewski
Role(s): UI & Game Flow Programmer
Primary responsibility for this work: Ensure the game works properly and design the instructions, end screen, and key functions to be easily understandable.

Goal of Work Session: Create a start screen and end screen, as well as a help screen.

Tools, Resources, or Inputs Used:

- Visual Studio Code
- GitHub

GenAI tools (if used)

ChatGPT 5.2

Date Used:

March 11 2026

Purpose of Use: GenAI was used to assist with creating the start and instructions page by providing code to help with basic layouts, templates, and inputs on best design choices.

Summary of Interaction: I uploaded the side quest 3 as an example since it had a start and end screen similar to what we needed, as well as our game synopsis. I then uploaded the necessary files from our game code such as our sketch.js file so that it could modify the code accordingly. GenAI instructed me on what to change, as well as what files to add such as the start.js and end.js files. After getting the basic and general layout of the code, I then kept refining it and asking GenAI to change the code based on my specific modifications and how I wanted the code to look like. I then worked on the key function instructions and asked GenAI what the best way to implement this to our game would be, as well as my own personal specifications for how I wanted certain parts to be.

Human Decision Point(s): I decided on most of the design aspects such as the key functions screen being separated by lines, having them look like keys on a keyboard, and it being a separate page from the main start page. I also decided to add the button on the top right if players need to refer to these instructions again. I also changed the layout and content of what GenAI provided such as spacing, text sizes, text content, etc. For the start page I decided to separate the goal section from the energy, low energy, and recover sections, and made the remaining three sections horizontal instead of vertical to make better use of space as they were initially overlapping and going off the screen.

Integrity & Verification Note: GenAI was used to assist with modifying previously written code into a base template for the game's start and end screens. All final code adjustments and integration into the repository were completed manually.

Scope of GenAI Use: GenAI did not contribute to the majority of the design of the code, as I put my own numbers, colors, and specifications of what I wanted GenAI to produce.

Limitations or Misfires: Sometimes GenAI would corrupt the code to where it would no longer work when trying to view the game, and I had to keep refining my prompts to receive a working outcome.

Verification & Judgement: The code was tested in visual studios to confirm that the start screen and end screens, as well as the game functions screen worked and loaded correctly. After testing, the code was reviewed and modified.

Appendix:
https://chatgpt.com/c/69b236c7-2634-8328-98fa-50532a0e34c0
https://chatgpt.com/c/69a7b75c-bfb8-832d-9473-8930ca146bdc

Entry Header 3

Name: Nila Sivapragasam

Role(s): Iteration Programmer

Primary responsibility for this work: Implementing the post-playtesting improvements

Goal of Work Session:

- Designed and implemented the maze layout
- Programmed the heart respawn mechanic

Tools, Resources, or Inputs Used:

- Visual Studio Code
- GitHub
- Lecture notes
- Microsoft Copilot

GenAI tools (if used)
Microsoft Copilot

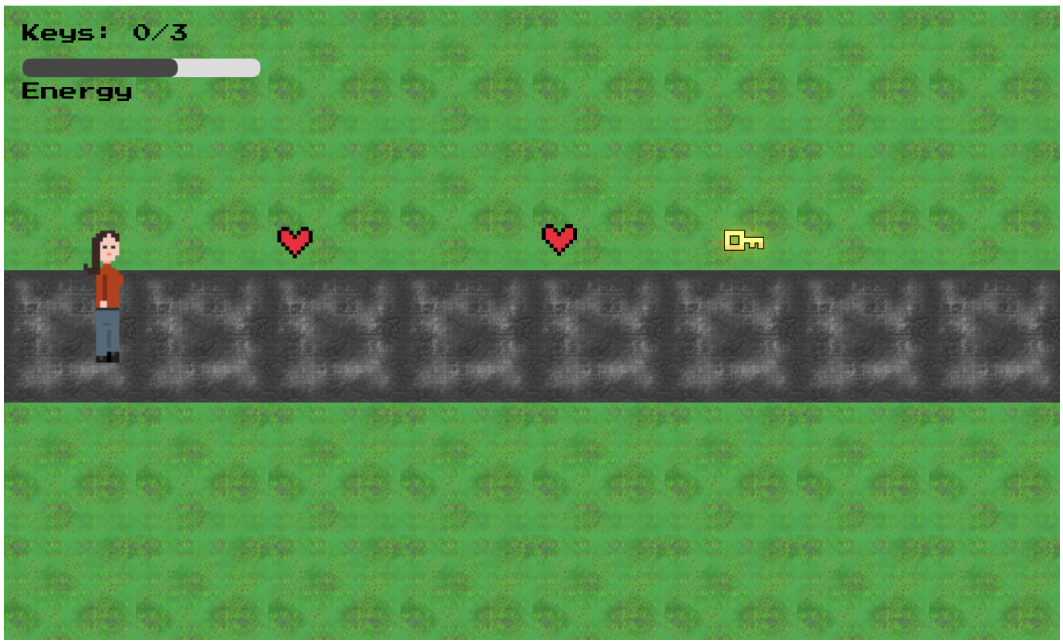
Date Used: March 11th 2026

Purpose of Use: To assist with implementing the maze layout and programming the heart respawn mechanic.

Summary of Interaction: I uploaded our current game code to Copilot and explained that I wanted the player to move through a maze rather than an open field. Copilot suggested a way to implement a maze structure within the existing level. After using this output, I prompted Copilot with additional requests to modify specific parts of the code, such as adjusting how quickly the player's energy depleted, changing where keys spawned, and modifying how the camera followed the player. I also asked Copilot to change the way the maze was implemented, because I wanted the maze to be generated from an array rather than drawing each wall manually.

Additionally, I asked how to implement a system where hearts would respawn 10 seconds after being collected.

Original game (before I added the maze):



Human Decision Point(s):

I decided to change how the maze layout was implemented. Copilot's original code drew each wall manually, which would make it difficult to modify the maze. After reviewing the Week 4 Example 03 code from class, I found that storing the maze in an array in the JSON file was a more efficient approach and would make it easier to create new levels in the future. I therefore asked Copilot to adjust the code so the maze would be generated from an array instead.

How Copilot originally drew the maze:

```
  "walls": [
    { "x": 0, "y": 300, "w": 2400, "h": 20 },
    { "x": 0, "y": 700, "w": 2400, "h": 20 },

    { "x": 200, "y": 500, "w": 1800, "h": 20 },
    { "x": 200, "y": 600, "w": 1800, "h": 20 },

    { "x": 600, "y": 320, "w": 20, "h": 380 },
    { "x": 1100, "y": 320, "w": 20, "h": 380 },
```

```
        { "x": 1600, "y": 320, "w": 20, "h": 380 },

        { "x": 850, "y": 500, "w": 120, "h": 20 },
        { "x": 1350, "y": 600, "w": 120, "h": 20 }
    ],
```

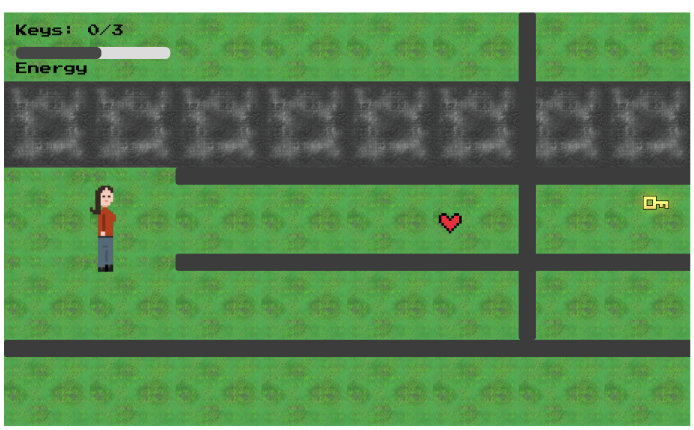
After implementing the array-driven maze, I modified aspects of the generated code, such as how the maze was drawn and where keys and hearts spawned in the level, to create a more challenging game. I also decided to make the player's energy deplete more slowly to ensure players still had the opportunity to find hearts before all their energy was gone.

Integrity & Verification Note: I carefully reviewed all GenAI outputs to ensure they were accurate and aligned with concepts covered in class. I read through the generated code to ensure I understood how it functioned and tested the implementation through gameplay to confirm that it worked correctly and supported our intended gameplay experience.

Scope of GenAI Use: GenAI was used to assist with generating and modifying specific gameplay code. I used Copilot to help implement the maze system, adjust mechanics, and create the heart respawn functionality. However, I manually edited the generated code to refine the maze layout, adjust where hearts and keys spawned, and slow down how quickly the player's energy depleted.

Limitations or Misfires: The initial maze generated by Copilot did not resemble a functional maze and instead appeared as randomly placed walls.

Original "maze" created by Copilot:



Additionally, after Copilot implemented the array-driven maze, it generated some keys on top of wall tiles, making them unreachable, so I had to manually change their location. Copilot also added comments to JSON files, which are not supported, so I had to remove those manually.

Appendix:
https://docs.google.com/document/d/114mxcZTEbOiw_H1pznzLpGcXb9ghM0waKTRkcoVAY64/edit?usp=sharing (placed in a separate document to avoid making this document too long).

Entry Header 4

Name: Cynthia Yang

Role(s): Level & Systems Integration Programmer + Readme & Process Documentation

Primary responsibility for this work: Level & Systems Integration Programmer + Readme & Process Documentation

Goal of Work Session: To refine the game level structure, ensure gameplay systems were properly integrated, and organize project documentation so that the code, design decisions, and development process were easier for others to understand and navigate.

Tools, Resources, or Inputs Used:

- Visual Studio Code
- GitHub

GenAI tools (if used)
ChatGPT 5.3

Date Used: March 10th & 11th 2026

Purpose of Use: ChatGPT was used as a support tool to help refine and organize the written documentation of the project, including improving clarity in the README file and development process notes. It was also used to review portions of the code and suggest ways to simplify structure and improve readability.

Summary of Interaction: I provided sections of the project documentation and portions of the game code to ChatGPT. The tool suggested clearer wording for explanations, improved organization of documentation sections, and provided recommendations for making the code

easier to follow. These suggestions included restructuring comments, clarifying variable purpose, and improving readability of certain code blocks.

Human Decision Point(s): All final decisions regarding documentation wording, code structure, and implementation were made by me. Suggestions from ChatGPT were reviewed and selectively adopted only when they aligned with the project's goals and accurately represented the work completed by the team.

Integrity & Verification Note: The outputs generated by ChatGPT were treated as suggestions rather than final content. I verified all changes to ensure they accurately reflected the project work and did not introduce incorrect information or unintended code behavior.

Scope of GenAI Use: ChatGPT was used primarily for documentation refinement, code readability improvements, and organizational support. It was not used to generate the core game mechanics or replace original development work.

Limitations or Misfires: Some suggestions from ChatGPT required adjustment because they did not fully match the existing structure of the project or the specific implementation details of the game. These suggestions were either modified or not used.

Verification & Judgement: All documentation and code changes were reviewed and tested after applying any suggested improvements. This ensured that the game functioned correctly and that the documentation accurately reflected the project's development process.

Appendix:

My prompt #1:

"organize this Project/Assignment Decisions 1. Maze Environment Design What changed: The game environment was redesigned into a maze structure with walls and pathways instead of a simple open environment. Players must navigate through the maze to reach the objective. Why the decision was made: The initial design lacked challenge and directional gameplay. Introducing a maze created a clearer game objective and spatial challenge, encouraging players to explore and think strategically about navigation. This also improved the visual structure of the game world and made the experience feel more complete. Who was involved: The maze concept was discussed collectively by the group during development. Layout ideas were proposed during group meetings, and the programming team implemented the maze structure, while other

members contributed suggestions for navigation flow and layout. 2. Heart Respawn Mechanic What changed: A heart-based respawn system was introduced. When a player loses, they are able to respawn using a heart instead of immediately ending the game. Why the decision was made: Early testing suggested that players found immediate failure frustrating. The heart system allows players to continue playing after mistakes, making the game more forgiving while still maintaining consequences for losing lives. This mechanic also improves engagement by encouraging players to keep exploring the maze. Who was involved: This idea came from group discussions after internal testing. The respawn logic was implemented by the developer responsible for gameplay mechanics, while the heart icon and visual representation were added as part of the design process. 3. Clearer Instructions Page What changed: The instructions page was redesigned to include clearer explanations of gameplay mechanics, controls, and objectives. Why the decision was made: During testing, some players were unsure how to begin playing or what the goal of the game was. Improving the instructions helped reduce confusion and made the game easier for first-time players to understand. Who was involved: This update was based on group feedback and playtesting observations. The text was rewritten collaboratively, and the page was integrated into the game interface by the developer."

ChatGPT 5.3 Output #1:

"Here is a **clean, structured, and rubric-aligned version** of your **Project/Assignment Decisions (Cynthia)** section. It keeps the required elements **What changed, Why, and Who**, but organizes them clearly in bullet form so it reads more professionally.

Project / Assignment Decisions (Cynthia)

1. Maze Environment Design

What changed

- The game environment was redesigned into a **maze structure with walls and pathways** rather than a simple open environment.

- Players now need to **navigate through the maze to reach the objective**, creating a more structured level layout.

Why the decision was made

- The original environment lacked challenge and clear direction.

- Introducing a maze created a **clearer gameplay objective and spatial challenge**.
- It encourages players to **explore the environment and think strategically about navigation**.
- The maze structure also improved the **overall visual organization of the game world**, making the level feel more complete.

Who was involved

- The concept of the maze was discussed **collectively by the group during development meetings**.
- Layout ideas were proposed by several team members.
- The **programming team implemented the maze structure**, while other members contributed suggestions to improve navigation flow and level design.

---

2. Heart Respawn Mechanic

What changed

- A **heart-based respawn system** was introduced.
- When the player loses, they can **respawn using a heart instead of immediately ending the game**.

Why the decision was made

- Early internal testing showed that **instant game over states felt frustrating** for players.
- The heart system allows players to **continue playing after making mistakes**, creating a more forgiving gameplay experience.
- It maintains consequences for failure while **encouraging players to keep exploring and improving their progress**.

Who was involved

- The idea emerged from **group discussions following playtesting feedback**.
- The **gameplay developer implemented the respawn logic**, while the **visual design elements for the heart icon** were developed during the design process.

---

3. Clearer Instructions Page

What changed

- The instructions page was **redesigned with clearer explanations of gameplay mechanics, controls, and objectives**.
- The layout and wording were improved so players can quickly understand how to start and play the game.

Why the decision was made

- Early testers experienced **confusion about controls and the overall objective of the game**.
- Improving the instructions helps **reduce onboarding friction for new players** and allows them to begin playing without external explanations.

Who was involved

- The decision came from **group feedback and playtesting observations**.
- The instructions text was **rewritten collaboratively by the team**, and the page was **implemented into the game interface by the developer responsible for system integration**.

If you want, I can also show you **how to format this so it fits perfectly inside your A1–A3 documentation page** (which professors often expect to be **shorter and more decision-focused**)."

My prompt #2:

“include the maze improvements, heart respawn and clearer instruction & control page”

ChatGPT 5.3 Output #2:

“Here is a **refined and organized version** that clearly highlights the **maze improvements, heart respawn system, and clearer instructions & controls page** while following the **A1–A3 decision format**.

Project / Assignment Decisions (Cynthia)

1. Maze Environment Improvements

What changed

- The game environment was redesigned into a **maze layout with walls and structured pathways** instead of an open environment.

- The maze layout was further refined to **improve navigation flow and player movement**, making paths clearer while still maintaining a sense of challenge.

Why the decision was made

- The initial environment lacked direction and did not provide enough challenge for players.

- Introducing a maze structure created a **clear gameplay objective and spatial puzzle element**.

- Refining the maze layout helped balance **difficulty and playability**, ensuring players could navigate the level without becoming overly confused or stuck.

Who was involved

- The maze concept and improvements were discussed during **group design meetings**.

- The **programming team implemented the maze structure**, while team members contributed ideas for layout adjustments based on gameplay testing.

---

2. Heart Respawn Mechanic

What changed

- A **heart-based respawn system** was added to represent player lives.
- When a player loses, they can **respawn using a heart instead of immediately ending the game**.

Why the decision was made

- Early testing showed that players became frustrated when the game ended immediately after failing.
- The heart system allows players to **continue playing after mistakes**, improving engagement and reducing frustration.
- It maintains consequences for failure while still encouraging players to **explore and learn the maze layout**.

Who was involved

- The idea came from **group discussions following internal playtesting**.
- The **gameplay developer implemented the respawn logic**, while the heart icon and visual elements were incorporated during the design process.

---

3. Clearer Instructions and Control Page

What changed

- The instructions page was redesigned to provide **clearer explanations of gameplay mechanics, controls, and objectives**.

- Control information and instructions were organized in a **more structured layout so players can quickly understand how to play**.

Why the decision was made

- Early testers were sometimes **uncertain about controls and the overall objective of the game**.

- Improving the instructions page helped **reduce confusion for new players and improve onboarding**.

Who was involved

- This change was based on **group feedback and playtesting observations**.

- The instructions text and layout were **rewritten collaboratively**, and the updated instructions page was integrated into the game interface during development.

This version is **clean, structured, and matches the assignment requirement** of documenting:

- What changed

- Why the decision was made

- Who was involved

while clearly emphasizing the **three major decisions: maze improvements, heart respawn, and clearer instructions/controls page**."

Erin O'Neill - A2 Process Documentation

Name: Erin O'Neill

Role(s): Environment & Asset Integration Programmer

Primary responsibility for this work: I created the assets and coded them to set up the game environment.

Goal of Work Session

The primary goal of the work I

Tools, Resources, or Inputs Used

- GenAI tools (if used): I used GenAI to write the code while I provided it with the ideas
- External references [1]

GenAI Documentation

Date Used: 2026-03-12

Tool Disclosure: ChatGPT 5.2

Purpose of Use: We used AI to devise the roles for the group character, as well as to write my code while I provided it with direction.

Summary of Interaction: First, I gave the AI a description of the game along with all of the file names and some of the primary code. First, I asked how to input my images I had already made into the already existing functions my teammates had made. Then from there I had to consistently correct the AI to get the outcome I wanted, which took a long time, Finally I added some highlights to the assets, like making the hearts and keys float and glow to promote interaction.

Human Decision Point(s): There were many times when the AI messed up. Part of the reason why I had to create a new chat was that during my previous chat, it began to get so long that it was glitching, and it wasn't understanding what I was trying to say anymore, so I started a fresh chat, which ended up working. For example, when I laid the tiles in the game, the AI originally laid them in a diagonal pattern; however, my initial vision was for it to create a path. Another example of one of the many decision points is the fact that I designed all of the assets in this game except for the font. I also made the assets, such as the heart and key, float and glow to show that you can interact with them.

Integrity & Verification Note: To verify, I conferred with my teammates, checked the A2 outline as well as the class outline to confirm the correct use of AI.

Scope of GenAI Use: In this assignment, I used GenAI to write almost all of my code while I provided it with all of the ideas.

Limitations or Misfires: The AI consistently messed up because I think there were so many messages that it actually got confused, and I would ask it to do something, and then it would tell me the fix for it, but then the fix actually changed nothing, or it just spat the same info out over and over again. Hence, I made a new chat.

Summary of Process (Human + Tool)

Overall, the AI helped me insert the drawings I had already made into code, which already existed, and was made by my teammates who were in charge of creating the controls and the base game. I also changed the formatting, colours, and style of the text, which my teammates had already created. Overall, my job was to fix the visuals of the game.

Decision Points & Trade-offs

Unfortunately, we did have to have some trade-offs due to limitations on time. We struggled with AI to get the code we already had. We had to sacrifice our storytelling screens because, for some reason, it would stop the character from moving.

Verification & Judgement

I felt it was appropriate given the class materials and contexts. Of course, all AI use was ensured to be within the class guidelines by comparison to the class syllabus.

Limitations, Dead Ends, or Open Questions

There were many dead ends, unfortunately. I wanted to do more with the game, such as adding a screen including a story which the character can follow along with, but unfortunately, adding that additional screen didn't seem to work. And because we were low on time, we had to just cut that completely and continue.

Appendix

Chat 1:
https://chatgpt.com/share/69b3038c-7880-800c-9ba5-a6d68a497775

Chat 2:
https://chatgpt.com/share/69b303bd-8b18-800c-b1db-d9d6495e4246